

Vishesh Kumar Singh

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About

I am a third-year undergraduate at the Indian Institute of Technology, Kanpur, pursuing a B.S. in Mathematics and Scientific Computing. My primary research interests include **Alignment in Large Language Models**, **Representation Learning**, and **Reinforcement Learning**. I have also worked with computational neuroscience as a secondary interest and **lead the Brain and Cognitive Science Club IITK**, which is working on it. I am currently pursuing undergraduate research under the guidance of **Prof. Sayak Ray Chowdhury** over test-time alignment in LLMs.

Education

Indian Institute of Technology(IIT) Kanpur

Expected May 2028

B.S. in Mathematics and Scientific Computing

Publications & Preprints

- **Mimicking Biological Memory Consolidation in SNNs** (Under Revision, Elsevier Neurocomputing)

Authors: Vishesh Kumar Singh

Simulated memory consolidation in SNNs via local engram identification to mimick long term potentiation in the biological brain, surpassing standard continual learning methods like EWC and SI adapted for SNNs by over 30%.

- **Dynamic Distractor Pruning in Deep RL (EMA-Tree SAC)** (Submitted to Springer ML)

Authors: Vishesh Kumar Singh

Developed EMA-Tree SAC using random forests and exponential moving averages to dynamically filter high-variance noise in continuous control, yielding a 51% return improvement.

Research Experience

Undergraduate Researcher

Jul 2026 – present

Advisor: Prof. Sayak Ray Chowdhury | Dept. of CSE, IIT Kanpur

- Working on test-time alignment focusing on mitigating hallucination and ensuring safety in Large Language Models.
- Investigating how steering the internal latent representations can lead to correct and safer response generation.

Research Intern (SURGE Program)

May 2026 – Jul 2026

Mentor: Prof. Sayak Ray Chowdhury | Dept. of CSE, IIT Kanpur | Preparing submission for ICLR 2027

- Exploited Best-of-N (BoN) strategies for test-time alignment in LLMs as an alternative to standard RLHF pipelines.
- Invented Reward Weighted Self Consistency improving performance over standard BoN by upto 10% and SPRT-RWSC for mathematically provable early stopping of candidates generation reducing budgets by up to 65%.

Reinforcement Learning Research Intern

Jun 2026 – Present

Mentor: Prof. Shakti Singh Gupta | Wadhvani School of AI & Intelligent Systems, IIT Kanpur

- Integrated missing NERO robot support in Robosuite for data collection to train baseline imitation learning policy.
- Implementing the Beyond Action Residuals framework in simulation using Robosuite for AgileX NERO arm.
- Handling the Sim-to-Real transition from trained simulation-based policy to the real NERO hardware.

Selected Projects

Spike Verse

May 2025 – Jul 2025

Brain and Cognitive Science Club IIT Kanpur

- Implemented Deep-RL algorithms like DQN, Double DQN, PER, Duelling Networks and N-Step Learning
- Showed 3x increase in the highest scores via weight transfer to a Spiking Neural Network from the trained ANN.

Closed Loop Discovery of Novel Fragrance Molecules

Ongoing

Brain and Cognitive Science Club IIT Kanpur

- Proposed the core idea to exploit the latent space of valid fragrance molecules for getting new valid compounds of desired fragrance by interpolation of known scents.
- Initial proof-of-concept experiments showed generation of 100% valid chemical compounds, of which 97% of the unique generated compounds proved to be novel, which were never seen in the training data of the JT-VAE

Major Competitions & Challenges

- **Samsung EnnovateX AX Hackathon 2026:** Amongst top 6 teams nationally qualified for pre-final round out of 1000+ participating teams from top national universities and colleges.
- **Inter IIT Techmeet 14.0:** Contributed as a contingent member to an overall second position of IIT Kanpur.
- **Build with Opus-4.7:** Got selected in the top 500 people of 20 K+ participants around the globe for round 2.

Scholastic Achievements

- Recieving INSPIRE SHE scholarship with an annual grant of INR 60K, awarded to top 1% of students by the Government of India.
- **JEE Advanced 2024:** Widely considered as second toughest exams in the world. Scored an All Indian Rank 786 out of 190K shortlisted candidates from JEE Mains 2024.
- **JEE Mains 2024:** Placed amongst top 0.013%, ranking 1904 out of 1.4 Million+ registered candidates, with a perfect score of 100 in the physics section.
- **National Standard Examination in Chemistry 2023:** National top 1% out of 44 K+ participants and was amongst selected candidates for INChO 2024.
- **National Standard Examination in Physics 2023:** Top 1% in state out of 4K+ participants and was placed amongst top 305 students qualified for INPHO 2024 out of 49K+ participants.
- Received an A* grade in the course Fundamentals of Computing, awarded only for very exceptional performance in course out of a class of 500+ students